



Table of Contents

Preface

Part 1: Introduction to WinUI and Windows Applications

1

Introduction to WinUI

Technical requirements

Before UWP – Windows 8 XAML applications

Windows application UI design

Windows Runtime (WinRT)

User backlash and the path forward to Windows 10

Windows 10 and UWP application development

Language choice with UWP development

Lifting app restrictions

UWP backward compatibility

What is XAML?

Creating an adaptive UI for any device

Powerful data binding

Styling your UI with XAML

Separating presentation from business logic

What is WinUI?

The first WinUI release

The road to WinUI 3

What's new in WinUI 3?

Goodbye UWP?

New features for WinUI 3 and the Windows App SDK

The Windows App SDK and WinUI

WinUI 3 compared to other Windows development frameworks

WinUI versus UWP

WinUI versus WPF

WinUI versus Windows Forms (WinForms)

Summary

Questions

2

Configuring the Development Environment and Creating the Project

Technical requirements

Installing Visual Studio and Windows desktop development workloads

Introducing the application idea

Reviewing the application features

WinUI in Desktop projects

Creating your first WinUI project

Anatomy of a WinUI in Desktop project

Reviewing App.xaml

Reviewing App.xaml.cs

Reviewing MainWindow.xaml

Reviewing MainWindow.xaml.cs

Reviewing the project references

Reviewing the project properties

XAML basics

Building the model

Creating sample data

Building the initial UI

Completing the data-binding initialization

Creating the DataTemplate and binding the UI

Understanding WinUI and Windows App SDK

Understanding the .NET app model

Working with WinUI controls, properties, and events

Adding a ListView header

Creating the ComboBox filter

Adding a new item button

Summary

Questions

3

MVVM for Maintainability and Testability

Technical requirements

Understanding MVVM

MVVM – the big picture

MVVM libraries for WinUI

The MVVM Toolkit

The Prism Library

MVVMCross

Choosing a framework for WinUI applications

Understanding data binding in WinUI

What are markup extensions?

Binding markup extension

x:Bind markup extension

Updating View data with INotifyPropertyChanged

Updating collection data with INotifyCollectionChanged

Implementing MVVM in WinUI applications

Working with events and commands

Implementing ICommand

Using commands in the View Model

Updating the View

Leveraging the MVVM Toolkit

Summary

Questions

4

Advanced MVVM Concepts

Technical requirements

Understanding the basics of DI

Using DI with ViewModel classes

Leveraging x:Bind with events

Page navigation with MVVM and DI

Migrating MainWindow to MainPage

Adding ItemDetailsPage

Adding new interfaces and services

Creating a navigation service

Creating a data service

Increasing maintainability by consuming services

Handling parameters in ItemDetailsPage

Creating the ItemDetailsViewModel class

Summary

Questions

5

Exploring WinUI Controls

Technical requirements

Understanding what WinUI offers developers

[Animated visual player \(Lottie\)](#)

[Navigation View](#)

[Parallax view](#)

[Rating control](#)

[Exploring the WinUI 3 Gallery app for Windows](#)

[Learning about the ScrollViewer control](#)

[Reviewing what's new in WinUI 3 and the Windows App SDK](#)

[Backward compatibility](#)

[Fluent UI and modern look and feel](#)

[Visual Studio tooling](#)

[The WebView2 control](#)

[What's new in the Windows App SDK](#)

[Adding some new controls to the project](#)

[Using the SplitButton control](#)

[Adding a TeachingTip control to the Save button](#)

[Summary](#)

[Questions](#)

[6](#)

Leveraging Data and Services

Technical requirements

Managing application state with app lifecycle events

Exploring Windows application lifecycle events

Lifecycle events of WinUI applications

Additional lifecycle events with FrameworkElement objects

Creating a SQLite data store

What is SQLite?

Adding SQLite as a data service

Leveraging a Micro ORM to simplify data access

Adding Dapper to the project

Updating the data service's initialization

Retrieving data via services

Summary

Questions

Part 2: Extending WinUI and Modernizing Applications

7

Fluent Design System for Windows Applications

Technical requirements

What is the Fluent Design System?

Exploring Fluent Design for Windows

Controls

Patterns

Layout

Input

Style

Incorporating Fluent Design in WinUI applications

Updating the title bar

Changing the style of MainPage

Changing the style of ItemDetailsPage

Using the Fluent XAML Theme Editor

Colors

Shapes

Acrylic material and the Fluent Design System

Use Mica in WinUI applications

[Incorporate Mica into My Media Collection](#)

[Design resources and toolkits for Fluent Design](#)

[Summary](#)

[Questions](#)

[8](#)

[Adding Windows Notifications to WinUI Applications](#)

[Technical requirements](#)

[Overview of push notifications in the Windows App SDK](#)

[Raw push notifications](#)

[Cloud-based app notifications](#)

[Local app notifications](#)

[Using raw push notifications in WinUI applications](#)

[Adding Windows app notifications with the Windows App SDK](#)

[Summary](#)

[Questions](#)

[9](#)

Enhancing Applications with Community Toolkits

Technical requirements

Introducing the WCT

Origins of the WCT

Reviewing recent toolkit releases

Exploring the WCT Gallery app

Installing and launching the app

Controls

Using controls from the toolkit

Creating the WinUI project

Referencing WCT packages

Adding data to the DataGrid control

Adding controls to the MainWindow control

Exploring the toolkit's helpers, services, and extensions

Helpers

Extensions

.NET Community Toolkit features

Summary

Questions

10

Accelerating App Development with Template Studio

Technical requirements

Overview of Template Studio for WinUI

Starting a new WinUI project with Template Studio

Exploring the code generated by Template Studio

Exploring the Core project

Exploring the main project

Exploring the MSTest project

Template Studio extensions for other UI frameworks

Template Studio for WPF

Template Studio for Uno Platform

Summary

Questions

Part 3: Build and Deploy on Windows and Beyond

11

Debugging WinUI Applications with Visual Studio

Technical requirements

Debugging in Visual Studio

Debugging local applications

Debugging remote applications

Common XAML layout mistakes

Improving your XAML with static code analysis

Pinpointing data binding failures

Common mistakes in data binding

Using the XAML Binding Failures window

Debugging live data with Live Visual Tree and Live Property Explorer

Coding with XAML Hot Reload

Debugging with Live Visual Tree and Live Property Explorer

Summary

Questions

12

Hosting a Blazor Application in WinUI

Technical requirements

Getting started with ASP.NET Core and Blazor

A brief history of ASP.NET and ASP.NET Core

What is Blazor?

WebAssembly and client-side .NET development

Creating a Blazor Wasm application

Building a simple application for tracking tasks

Exploring Blazor Wasm deployment options

Deployment options for Blazor Wasm projects

Publishing Blazor to Azure Static Web Apps hosting

Pushing the project to GitHub

Creating an Azure Static Web Apps resource

Publishing an application with GitHub Actions

Hosting your Blazor application in the WinUI WebView2

Summary

Questions

13

Take Your App Cross-Platform with Uno Platform

Technical requirements

An overview of Uno Platform

Creating your first Uno Platform project

Migrating WinUI XAML markup and code to Uno Platform

Migrating the WinUI project code

Migrating the WinUI XAML views

Running on Android with WSA

Running in the browser with WebAssembly

Summary

Questions

14

Packaging and Deploying WinUI Applications

Technical requirements

Discovering application packaging and MSIX basics

MSIX

Reviewing MSIX tools and resources

Packaged applications and application identity

Getting started with application packaging in Visual Studio

Deploying applications with Windows Package Manager

Adding a package to the community repository

Using WinGet for package management

Distributing applications with the Microsoft Store

Preparing a free application for the Microsoft Store

Uploading a package to the Store

Sideloaded WinUI applications with MSIX

Creating an MSIX package for sideloading

Sideloaded an MSIX package

Summary

Questions

Index

Other Books You May Enjoy